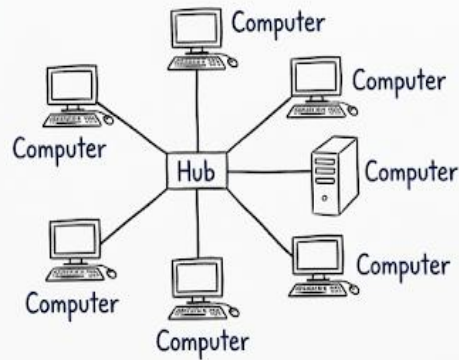


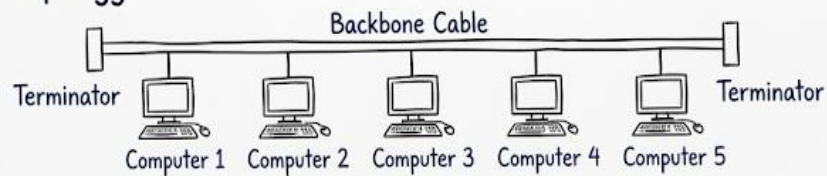
Network Topologies

1. Image of Topologies :

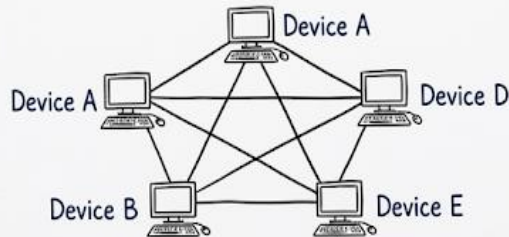
Star Topology



Bus Topology



Mesh Topology



2. Topology Comparison :

Topology	Pros (Advantages)	Cons (Disadvantages)
Star	• Easy to add new devices. • Single node failure doesn't stop the network. • Simple troubleshooting.	• If the central hub fails, the whole network dies. • Requires more cable length than bus.

Topology	Pros (Advantages)	Cons (Disadvantages)
	• High traffic performance.	• Higher setup cost due to switches/hubs. • Limited ports on central switch.
Bus	• Very low cable requirement. • Cheapest topology to implement. • Simple for small temporary setups. • No extra switch hardware required.	• Heavy traffic slows down the network. • Break in main cable downs entire network. • Difficult to find individual faults. • Low security (data passes every node).
Ring	• Equal data access for all nodes. • Performs better than bus under heavy load. • No need for a central server. • Orderly data traffic (token ring).	• One broken cable/node breaks entire loop. • Slower data transfer (travels sequentially).
Mesh	• Ultra-high reliability and redundancy.	• Extremely expensive due to extensive cabling. • Very complex to install and manage.

Topology	Pros (Advantages)	Cons (Disadvantages)
	• Simultaneous multi-path data transfer. • High privacy and data security. • No bottleneck issues.	• Redundant hardware overhead. • Uses large power and space resources.

3. Game server Architecture :

- **Chosen Topology: Hybrid Star-Mesh Topology**
- **Justification:** A central Star topology is used at the player level where regional client devices connect directly to central Matchmaking/Game Servers, while the underlying Game Servers are interconnected in a **Full/Partial Mesh Topology**.
- **Speed Impact:** Direct client-to-server connections minimize ping, while mesh server connections handle fast real-time synchronization between match lobbies.
- **Reliability Impact:** Extremely high; if one server node crashes mid-game, mesh routing instantly migrates the match state to an active fallback server without dropping players.
- **Cost Impact:** Higher initial infrastructure and server bandwidth costs, but essential for competitive low-latency gameplay.

4. Warehouse Network Topology :

